Oil Spill Detection in SAR Images using Transformer-based Architectures

Melisa Ünsalan
Doctoral School SD-ETTI,
NUST POLITEHNICA Bucharest
Bucharest, Romania
melisa.unsalan@upb.ro

Anamaria Radoi
Department of Applied Electronics
and Information Engineering,
NUST POLITEHNICA Bucharest
Bucharest, Romania
anamaria.radoi@upb.ro

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***Abstract*—Oil spills are a serious environmental hazard, causing severe damage to marine ecosystems and negatively affecting coastal populations. In order to ensure a proper response, rapid and accurate detection is crucial. In this regard, many machine learning techniques have been proposed to address the problem of oil spill detection from images captured by SAR sensors, which are not affected by weather conditions or day/night illumination. The main challenges are the presence of speckle noise in SAR imagery and events that share a similar spectral signature to oil spills. In this paper, two approaches that are popular in computer vision are evaluated based on their ability to detect oil spills, namely U-Net, a convolutional network, and SegFormer, a Transformer-based architecture. The experiments prove that transformer-based methods lead to superior performance, i.e., 0.707 mIoU and 81.48 % accuracy reached for the B3 variant of SegFormer.**

***Index Terms*—oil spill, detection, segmentation, SAR, U-Net, SegFormer**

I. Introduction

Due to the ever-increasing oil demand, oil spills pose a significant threat to marine and coastal ecosystems. The most common causes of oil spills are human error, accounting for almost half of the cases, followed by equipment malfunction [1]. Such an event can have a devastating impact on wildlife, both short term and long term effects on the economy, and human health, impacted both by direct exposure to the oil and indirectly by eating contaminated seafood [2]. Thus, early detection of oil spills is important in order to ensure a rapid response. Remote sensing is a valuable tool for the detection and monitoring of the oil slicks as it can quickly provide vast amounts of data from areas difficult to access otherwise.

The main types of sensors used to acquire satellite data for oil spill detection are multispectral and microwave [3]. Multispectral sensors are passive and can acquire data from the ultraviolet to the near-infrared range of the electromagnetic spectrum. However, since they need daylight to operate and are severely hindered by cloud coverage, they are not used as commonly as microwave sensors. Radar systems are active since they transmit microwaves and measure the backscatter. The two most frequently used imaging systems from this category are synthetic aperture radar (SAR) and side-looking airborne radar (SLAR). On account of the fact that oil in the water dampens waves such as sea surface capillaries and short gravity waves, the intensity of the backscattered energy is reduced, thus, oil spills appear dark in SAR images [4]. Nonetheless, such spectral signatures can appear from a variety of causes, such as natural surface films from plankton or grease, algae, or low-wind areas.

Given that manually annotating remotely sensed images is a laborious task as they cover extensive areas, various automatic approaches have been proposed. Machine learning techniques are very popular for this task, given their strong generalization capabilities. Unfortunately, they require vast amounts of annotated data and computational resources for training.

Support Vector Machines (SVM) have shown good results with relatively few training samples, as shown in [5]. The features extracted are common features used in literature and an SVM was used to differentiate between the oil spill and look-alike classes. With active learning it was proven that the size of a dataset can be significantly reduced without hindering the performance if the proper strategy to select the samples is used.

An approach that combines the feature extraction abilities of deep neural networks with the robustness of SVMs has been developed, namely DRSNet [6]. A ResNet-50 [7] backbone is used to create embeddings for ten polarimetric features, followed by an Atrous Spatial Pyramid Pooling (ASPP) [8] module in order to improve multi-scale generalization. The final binary segmentation is derived by the SVM from the resulting feature maps.

Encoder-decoder structures are widely adopted in semantic segmentation tasks due to their ability to leverage the extraction of strong hierarchical features and reconstruct high-resolution segmentation maps based on the embeddings [9].

One of the most popular architectures employing this structure is U-Net [10] and its variants, which showed good results for the task of oil spill segmentation. One example is shown in [11], in which a U-Net with a ResNet-101 [7] backbone is used to test a new benchmark dataset, which is also used for

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the experiments presented in this paper.

In [12], an architecture called OilSeg-SARNet is proposed, building upon the U-Net architecture by incorporating components that enhance local feature correlations, retrieve multi-scale contextual information, and refine boundary details. Another enhanced version of U-Net is presented in [13], making use of nested skip connections (similar to U-Net++ [14]), multi-resolution convolutional blocks, and attention gates,

TransUNet [15] shows that combining U-Net with Transformers can yield better results than on their own. The model employs the standard U-Net architecture but adds a Transformer block following the CNN encoder, therefore leveraging the global scene-level reasoning capabilities of Transformers and the local fine-grained segmentation of the U-Net architecture.

The popular encoder-decoder structure is also adopted by CBDNet [16], [17], which employs a ResNet-34 [7] for feature extraction. After each residual block, a spatial and channel squeeze excitation (scSE) block is introduced to further improve the representation capabilities of the feature extractor. Subsequently, a dilated convolution module with different rates performs feature fusion across different scales, and the decoder reconstructs the final segmentation map.

ConvNeXt, a modern reinterpretation of the classical CNNs, is used as a backbone for [18], while also employing an attention mechanism that considers both spatial and frequency information. A Strip-Augmented Pyramid Pooling (SAPP) is introduced in the bottleneck of the network with the purpose of preserving the geometry of elongated oil spills.

Several transformer-based architectures have been proposed in the last few years. For example, the Swin transformer [19] was adopted in [20] and enhanced with a block named Multi-Scale Feature Selection and Fusion (MSFS) and a variant of MSFS suited for introducing global context. The traditional skip connections are replaced with a Multi-scale Feature Connection (MFC) block that enables the model to extract multi-scale embeddings.

Most of the transformer-based architectures require large amounts of annotated data to ensure proper training. Taking into account the limited amount of annotated datasets available for oil spill segmentation, a framework called TransOilSeg has been proposed in [21]. This approach heavily relies on transfer learning, using large scale datasets to pretrain the model in order to learn common features of data, followed by fine-tuning on oil spill datasets using a gradient aggregation technique.

The Segment Anything Model (SAM) [22] was also used for this purpose in [23] in conjunction with YoloV8, which supplies the bounding boxes used as prompts. The final architecture is called SAM-OIL. Since the original SAM is trained on optical images and too large for fine-tuning, a small trainable Adapter module is added on top of a frozen SAM, improving performance to a great extent.

II. Dataset

Due to the lack of publicly available datasets that include mask labels for oil spills and look-alikes, the dataset presented in [11], [24] was developed with the purpose of becoming a benchmark for oil spill detection tasks. In order to build a reliable ground truth, the timestamps and coordinates of oil spill events confirmed by the European Maritime Safety Agency (EMSA) between 28 September 2015 and 31 October 2017 are used to select the satellite image tiles.

The image data is acquired by Sentinel-1 satellites with a C-band SAR sensor operating in the Interferometric Wide Swath (IW) acquisition mode, resulting in a Ground Range Detected (GRD) product. Although the radar image is dual-polarized (VV + VH), the dataset is constructed using only the VV band. The pre-processing steps followed in order to obtain a dataset sample are: the oil spill is located from the EMSA records; the oil spill and surrounding context is cropped and rescaled to a 1250×650 pixel resolution; radiometric calibration is applied to the cropped and rescaled patch; a 7×7 median filter is used to reduce the speckle noise and a linear transformation is applied to the image dB values.

A total of 1112 images were extracted following this process which are subsequently split into training and testing subsets with a 90% - 10 % ratio, resulting in 1002 and 110 images, respectively. From the images, 5 classes are considered of interest: sea surface, oil spills, look-alikes, ships, and land. The classes are strongly imbalanced, with the sea surface or land classes occupying most of the image. In addition, look-alikes usually cover a larger area than oil slicks, adding to the imbalance. The ”ship” samples cover a very small area and are not found in every image.

III. Proposed Method

A. U-Net

The U-Net [10] architecture was originally developed for biomedical image segmentation as an extension of Fully Convolutional Networks (FCN) [25] that can perform pixel-wise segmentation with limited training data.

The U-Net architecture consists of a contracting path (encoder) and an expansive path (decoder). The contracting part follows the FCN architecture, being comprised of blocks that employ two 3×3 convolutions, each followed by a rectified linear unit (ReLU), and a 2×2 max pooling operation with stride 2 for downsampling. Each block uses double the number of feature maps of the previous one. The expansive path mirrors the encoder, also consisting of four blocks that begin with a 2×2 deconvolution layer for upsampling and two 3×3 convolutions, each followed by a ReLU. Between each encoder and decoder block, a skip connection exists in order to preserve fine-grained spatial details and boundaries that might be lost during the downsampling. Finally, the desired segmentation map is derived after a 1×1 convolution maps the embeddings to the desired number of classes.

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B. Transformer-based Models

Although transformer-based architectures were first used for natural language processing (NLP), they have achieved remarkable results in numerous computer vision tasks. The Vision Transformer (ViT) [26] obtained state-of-the-art results in image classification, but had two main disadvantages: it was computationally expensive and extracted single-scale feature maps. In order to avoid these limitations, SegFormer [27] was developed for semantic segmentation tasks.

Drawing inspiration from both ViTs and classical convolutional neural networks (CNNs), SegFormer features a hierarchical transformer encoder for multi-scale feature extraction and an MLP-based decoder for lightweight feature fusion. Being able to extract both low resolution coarse features and high resolution delicate features is very important since relevant features are present across multiple scales when dealing with complex phenomena.

SegFormer is comprised of a Mix Transformer encoder (MiT) that creates embeddings at four different resolutions, a simple feed-forward network (Mix-FFN) that preserves positional information instead of the classical transformer positional encoding in order to ensure performance across multiple scales, a lightweight self-attention mechanism, and a simple four step MLP decoder that aggregates extracted features and yields the final segmentation mask logits.

C. Joint Loss Function

Due to the nature of the task, classes are often strongly imbalanced, the background class (sea surface) accounting for almost 90% of the pixels in the dataset [11]. Hence, it is important to choose a loss function that takes into account the imbalance factor.

Cross-Entropy loss ($\mathcal{L}_{ce}$) handles each pixel independently, allowing for the preservation of fine-grained details [28]. However, it is not recommended when classes are imbalanced, as it will favor the background class. Dice loss ($\mathcal{L}_{dice}$) started gaining widespread use for segmentation tasks with imbalanced classes [20], [29], [30]. It evaluates the overlap between the segmentation mask returned by the model and the ground truth mask [28], being suitable for tasks where objects are small and scattered throughout the image.

We opt for the following custom loss function, similar to the one used in [15], [20], [20], [30]:

$$\mathcal{L}_{total} = \lambda \cdot \mathcal{L}_{ce} + (1 - \lambda) \cdot \mathcal{L}_{dice} \tag{1}$$

where λ is a coefficient that weights the stability of the cross-entropy loss and attention to small objects of the dice loss.

IV. Results

SegFormer [27] is provided in six variants (i.e., B0-B5) of varying model size, run-time, and accuracy, which depend on the scaling of the MiX Transformer (MiT) encoder. In order to observe the impact of the MiT encoder scaling on oil spill detection, we conducted experiments for all six variants of the SegFormer model. The best performance, relative to the number of parameters, was reached by the B2 and B3 variants of the SegFormer model, as shown in Table I. The performance is reported over the following standard metrics: overall pixel-wise accuracy, precision, and mean intersection over Union (mIoU).

TABLE I: Number of parameters of SegFormer models

SegFormer model	Number of parameters	Accuracy [%]	Precision [%]	mIoU
B0	3.7M	81.30	78.18	0.687
B1	13.6M	81.75	76.16	0.674
B2	27.3M	81.46	79.34	0.698
B3	47.2M	81.48	80.81	0.707
B4	63.9M	81.50	79.56	0.700
B5	84.5M	82.60	80.48	0.712

TABLE II: The impact of λ in the joint loss function

Method	λ	Accuracy [%]	Precision [%]	mIoU
SegFormer (B3)	0.25	83.68	78.89	0.707
	0.5	81.48	80.81	0.707
	0.75	83.53	79.15	0.706

The encoder of the SegFormer architecture was initially pretrained on Imagenet-1k [31], and a randomly initialized decoder. The model reaches convergence in a few number of epochs, i.e., less than 50 epochs, using an AdamW optimizer [32] and a learning rate of $lr = 5 \cdot 10^{-5}$. In order to investigate the impact of λ in the joint loss function, we train for $\lambda \in \{0.25, 0.5, 0.75\}$. The obtained results are provided in Table II. For $\lambda = 0.5$, a slightly lower accuracy is achieved. However, the precision and also the training convergence is improved if compared to the other values for λ. It is worth mentioning that the mean IoU is not greatly influenced by the choice of λ.

A comparison against other methods in terms of the IoU per class is provided in Table III. For almost all classes, the SegFormer-based model outperforms other methods. In the case of the ship class, the reduced number of examples in the training dataset leads to a smaller performance if compared to SAM-OIL [23]. This is expected since transformer-based models usually require a larger number of examples to obtain satisfactory results. The best IoU is achieved for the Sea Surface and Land classes, the two largest categories in the dataset, accounting for 93% of all pixels in the dataset. It is worth mentioning that the SegFormer B3 variant achieves a higher per-class IoU value, using a smaller number of parameters if compared to SAM-OIL or YOLOv8-SAM methods [23].

Some simple segmentation cases are shown in Fig. 1 and some of the more complicated ones in Fig. 2. The oil spill and look-alike classes have very similar spectral signatures and can be easily confused, as it can be observed from the first three rows from Fig. 2. Another challenge arises from the fact that depending on the wind conditions, the spills might be dispersed, resulting in difficulties in detecting the boundaries, such as in the last three rows from Fig. 2. The lowest performance is obtained for the ship class. This behavior is

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TABLE III: Comparison with other methods in terms of IoU per class [%]

Method	Parameters	Sea Surface	Oil Spill	Look-alike	Ship	Land
U-Net [11]	31 M	93.90	53.79	39.55	44.93	92.68
DeepLabV3 [11]	2.1 M	96.43	53.38	55.40	27.63	92.44
YOLOv8-SAM [23]	68.16 M	94.34	41.84	48.15	52.48	87.65
SAM-OIL [23]	68.16 M	96.05	51.60	55.60	**52.55**	91.81
SAGPNet [30]	-	92.28	46.04	57.39	-	88.67
MGPNet [20]	-	92.78	54.64	58.46	-	95.48
SegFormer (B3)	47.2 M	**96.61**	**57.85**	**59.42**	44.05	**95.65**

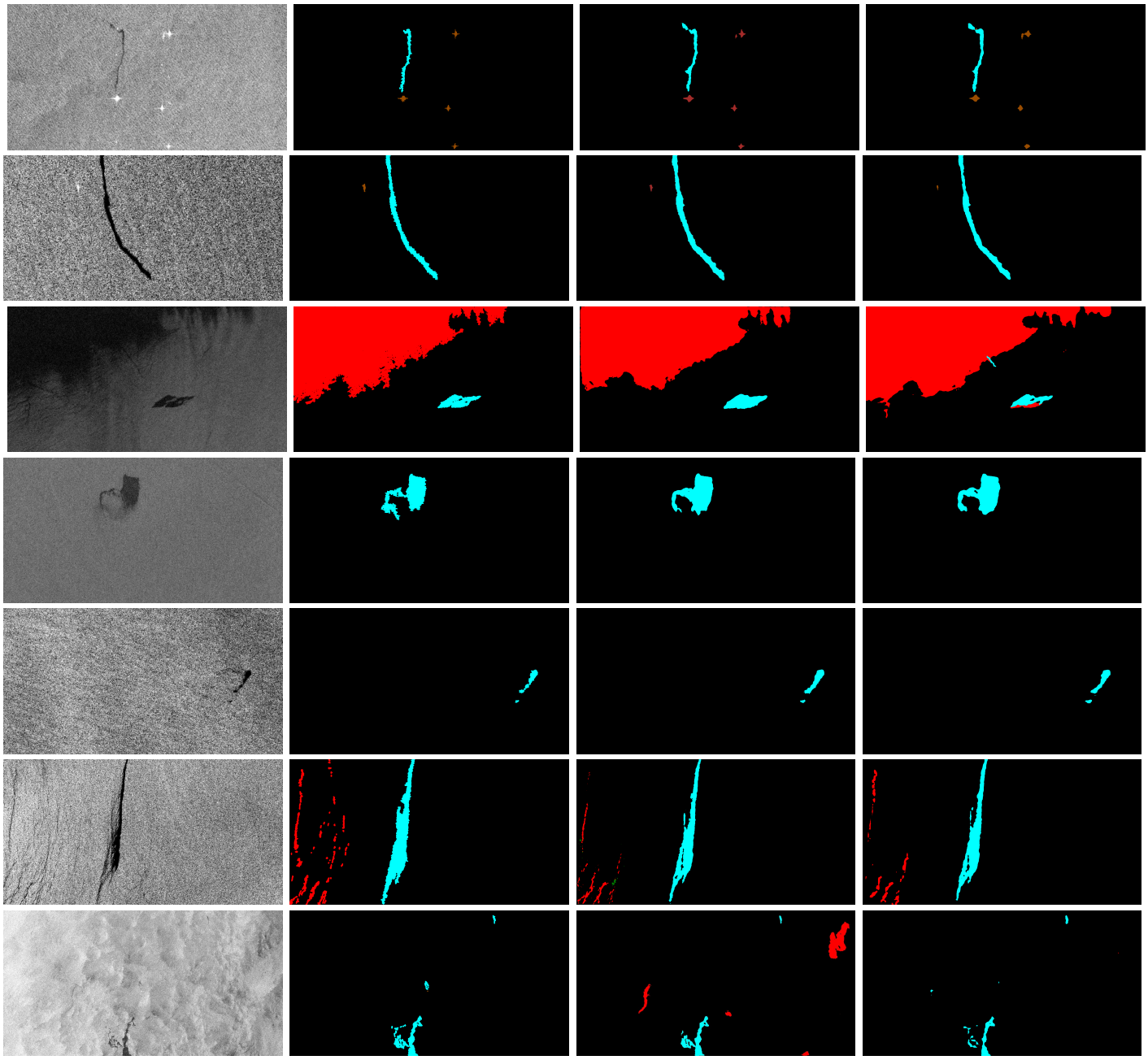


Fig. 1: Left to right: Original SAR image, ground truth, U-Net output, and SegFormer output. The following color legend is used: black - sea surface, cyan - oil spill, red - look-alike, brown - ship, and green - land.

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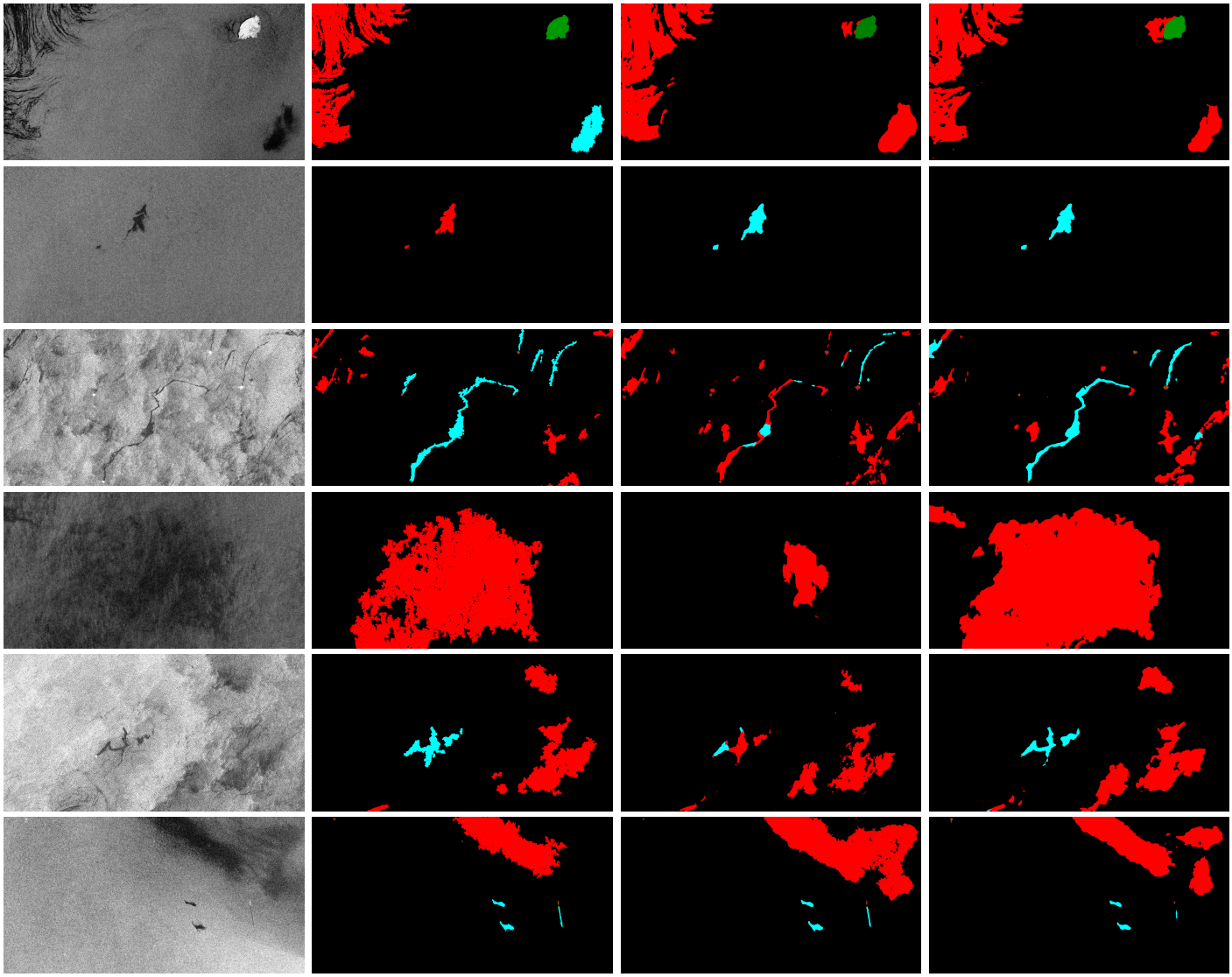


Fig. 2: Left to right: Original SAR image, ground truth, U-Net output, and SegFormer output. The following color legend is used: black - sea surface, cyan - oil spill, red - look-alike, brown - ship, and green - land.

expected since the ship class accounts for almost 0.03% of the pixels, and the objects in the ship class are much smaller in size compared to objects from other classes. Nevertheless, the correct identification of ships can provide important clues regarding the origin of the oil spill, i.e., its presence close to a streaky black spot most likely signifies an oil discharge from the ship, as in the first two rows from Fig. 1.

V. Conclusions

This paper highlights the effectiveness of the U-Net and SegFormer architectures for oil spill detection using SAR imagery. The fast and accurate detection of oil spills is critical in order to properly assess the environmental damage and ensure rapid response, but it can be difficult due to the speckle noise characteristic to SAR images and other phenomena that can produce similar spectral signatures. The experimental results show that both networks have strong oil spill segmentation capabilities, highlighting the effectiveness of encoder-decoder architectures due to their ability to extract both low-level and high-level embeddings and reconstruct segmentation maps that preserve fine details. Directions for future work include expanding the datasets with more relevant samples, especially highlighting the difference between oil spills and look-alikes, integrating multi-sensor data, and improving the models to ensure boundary data is well preserved and embeddings that help better distinguish between the classes are extracted.

References

[1] J. Michel and M. Fingas, "Oil spills: Causes, consequences, prevention, and countermeasures," in *Fossil Fuels: Current Status and Future Directions*. World Scientific, 2016, pp. 159–201.

[2] S. E. Chang, J. Stone, K. Demes, and M. Piscitelli, "Consequences of oil spills: A review and framework for informing planning," *Ecology and Society*, vol. 19, no. 2, p. 26, 2014.

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[3] R. Al-Ruzouq, M. B. A. Gibril, A. Shanableh, A. Kais, O. Hamed, S. Al-Mansoori, and M. A. Khalil, "Sensors, features, and machine learning for oil spill detection and monitoring: A review," *Remote Sensing*, vol. 12, no. 20, p. 3338, 2020.

[4] F. Girard-Ardhuin, G. Mercier, and R. Garello, "Oil slick detection by SAR imagery: potential and limitation," in *Oceans 2003. Celebrating the Past... Teaming Toward the Future (IEEE Cat. No. 03CH37492)*, vol. 4. IEEE, 2003, pp. 2026–2031.

[5] Y. Cao, L. Xu, and D. Clausi, "Active learning for identifying marine oil spills using 10-year RADARSAT data," in *2016 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2016, pp. 7722–7725.

[6] D. Wang, S. Liu, C. Zhang, M. Xu, J. Yang, M. Yasir, and J. Wan, "An improved semantic segmentation model based on SVM for marine oil spill detection using sar image," *Marine pollution bulletin*, vol. 192, 05 2023.

[7] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 770–778.

[8] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs," in *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2017.

[9] M. Jamshidi, A. Kalhor, and A. Vahabie, "Efficient compression of encoder-decoder models for semantic segmentation using the separation index," *Scientific Reports*, vol. 15, 07 2025.

[10] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2015, pp. 234–241.

[11] M. Krestenitis, G. Orfanidis, K. Ioannidis, K. Avgerinakis, S. Vrochidis, and I. Kompatsiaris, "Oil spill identification from satellite images using deep neural networks," *Remote Sensing*, vol. 11, no. 15, p. 1762, 2019.

[12] X. Lyu, H. Su, C. Grecos, and P. Ren, "A novel framework for marine oil spill detection in SAR imagery fusing edge supervision enhancement and group attention mechanism," *Remote Sensing Applications: Society and Environment*, vol. 41, p. 101901, 2026.

[13] M. Elsheref, P. J. Thapa, A. Katwal, A. Bin Naeem, M. A. Tarr, and M. T. Hoque, "OilSpillNet: A deep learning framework for accurate detection and segmentation of marine oil spills from imagery," *Journal of Environmental Chemical Engineering*, vol. 14, no. 2, p. 121714, 2026. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2213343726006883

[14] Z. Zhou, M. M. Rahman Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested u-net architecture for medical image segmentation," in *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support - 4th International Workshop, DLMIA 2018 and 8th International Workshop, ML-CDS 2018 Held in Conjunction with MICCAI 2018*. Springer Verlag, 2018, pp. 3–11.

[15] X. Dong, J. Li, B. Li, Y. Jin, and S. Miao, "Marine oil spill detection from low-quality SAR remote sensing images," *Journal of Marine Science and Engineering*, vol. 11, no. 8, 2023. [Online]. Available: https://www.mdpi.com/2077-1312/11/8/1552

[16] Y. Zhang, Q. Zhu, and Q. Guan, "Oil spill detection based on CBD-Net using marine SAR image," in *2021 IEEE International Geoscience and Remote Sensing Symposium IGARSS*, 2021, pp. 3495–3498.

[17] Q. Zhu, Y. Zhang, Z. Li, X. Yan, Q. Guan, Y. Zhong, L. Zhang, and D. Li, "Oil spill contextual and boundary-supervised detection network based on marine SAR images," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–10, 2022.

[18] S. Wang, B.-W. Min, D. Gao, and Y. Hong, "FCS-Net: A frequency-spatial coordinate and strip-augmented network for sar oil spill segmentation," *Journal of Marine Science and Engineering*, vol. 14, no. 2, 2026.

[19] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *IEEE International Conference on Computer Vision*, 2021.

[20] S. Dong and J. Feng, "MGPNet: Rethinking multi-scale features and global attention with pre-trained model for sar oil spill detection," *Marine pollution bulletin*, vol. 223, p. 119011, 11 2025.

[21] Y. Chai, X. Han, Y. Wang, D. Luo, J. Yang, P. Chen, and G. Zheng, "TransOilSeg: A novel SAR oil spill detection method addressing data limitations and look-alike confusions," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 63, pp. 1–16, 2025.

[22] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollar, and R. Girshick, "Segment anything," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, October 2023, pp. 4015–4026.

[23] W. Wu, M. S. Wong, X. Yu, G. Shi, C. Y. T. Kwok, and K. Zou, "Compositional oil spill detection based on object detector and adapted segment anything model from SAR images," *IEEE Geoscience and Remote Sensing Letters*, vol. 21, pp. 1–5, 2024.

[24] M. Krestenitis, G. Orfanidis, K. Ioannidis, K. Avgerinakis, S. Vrochidis, and I. Kompatsiaris, "Early identification of oil spills in satellite images using deep cnns," 01 2019, pp. 424–435.

[25] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *IEEE Conference on Computer Vision and Pattern Recognition*, 2015, pp. 3431–3440.

[26] A. Kolesnikov, A. Dosovitskiy, D. Weissenborn, G. Heigold, J. Uszkoreit, L. Beyer, M. Minderer, M. Dehghani, N. Houlsby, S. Gelly, T. Unterthiner, and X. Zhai, "An image is worth 16x16 words: Transformers for image recognition at scale," 2021.

[27] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[28] R. Azad *et al.*, "Loss functions in the era of semantic segmentation: A survey and outlook," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2023.

[29] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Fourth International Conference on 3D Vision*, 2016, pp. 565–571.

[30] S. Dong and J. Feng, "SAGPNet: A shape-aware and adaptive strip self-attention guided progressive network for sar marine oil spill detection," *Marine Environmental Research*, vol. 204, p. 106904, 2025.

[31] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2009, pp. 248–255.

[32] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," *International Conference on Learning Representations*, 2019.

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